Contribution Results

Reliable Hybrid SLC/MLC ReRAM Crossbars for Transformers

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Overview

The baseline is the HyFlexPIM hybrid (SVD components ranked by the loss gradient, 20% in SLC), with all-SLC and all-MLC chips as references (see *Baseline Results*). We contribute five things, in the order of the report: (1) a verified cell-level crossbar model and the finding that stuck cells are the limiting effect; (2) a Fisher-information estimate of each component's damage, which shows that ranking by singular value (magnitude) is close to the best placement; (3) fault-aware mapping; (4) tier-aware training; (5) evidence that (2) and (3) hold on MNLI. Every method is tested on the same simulated chips as the baseline, so differences within an experiment are paired (mean ± standard error, in accuracy points unless stated). The hybrid chip uses 9.2 cells per weight, 34% fewer than all-SLC.

1 Contribution 1: cell-level crossbar model and characterisation

What. A simulator that stores every weight as a real crossbar would: 8-bit integers with one scale per output column, a differential cell pair for the sign, and 7 SLC or 4 MLC bit slices per sign. Errors are applied to every cell: programming noise, read noise, level-dependent MLC noise, stuck cells in three spatial patterns, IR drop, and a fixed ADC with bit-serial inputs. Self-tests show that, with all errors off, it equals exact 8-bit integer arithmetic (relative error $< 10^{-4}$), and an error-free chip stays within about one point of the digital model. HyFlexPIM instead perturbs whole weights, $\tilde{W} = W \odot (1 + \eta)$, at a single noise level, so these effects cannot be studied there.

Table 1: Our placements under each non-ideality, one at a time, with 20% SLC: accuracy (%) of magnitude and Fisher, and the difference of magnitude to the HyFlexPIM gradient hybrid and to all-SLC on the same chips (baseline values in the *Baseline Results*; complete sweeps in Appendix A.4).

	BERT-Base				DistilBERT			
Condition	Magnitude	Fisher	vs HyFlexPIM	vs all-SLC	Magnitude	Fisher	vs HyFlexPIM	vs all-SLC
No errors	90.6	89.6	+0.3	+0.9	88.8	89.1	+0.1	+0.2
Read noise $\sigma_r = 2\%$	90.1	89.7	+0.3	+1.1	88.4	88.2	+0.3	+0.1
Read noise $\sigma_r = 5\%$	82.5	82.7	+5.8	+2.5	83.3	83.2	+2.9	-1.2
$\sigma = 5\%$, middle MLC levels $\times 2$	88.4	88.4	+0.5	+0.2	88.0	87.9	+1.0	-0.4
ADC 6 bits (bit-serial)	90.0	90.5	+0.5	+0.7	89.7	88.8	+1.3	+1.7
ADC 5 bits (bit-serial)	51.1	51.6	-1.8	+0.0	70.6	70.9	-6.7	-0.3
IR drop $r = 3 \times 10^{-4}$	90.4	91.1	-0.1	+1.5	89.7	89.3	+0.1	+0.2
IR drop $r = 3 \times 10^{-3}$	76.8	72.8	+0.5	+18.3	73.4	72.6	-1.3	+3.3
1% stuck cells, scattered	56.4	55.6	+3.8	-5.0	62.5	60.0	+7.5	-11.7
1% stuck cells, 3×3 clusters	54.0	53.1	+2.5	-12.7	56.1	57.3	-0.3	-24.1
1% stuck cells, bit lines	50.4	52.2	-0.6	-6.6	54.7	56.3	+2.5	-16.5

Takeaways.

- Under read noise, level-dependent noise, a 6-bit ADC and moderate IR drop, the magnitude and Fisher hybrids stay within about a point of all-SLC or above it, and magnitude is 2.9–5.8 points ahead of the HyFlexPIM hybrid at 5% read noise. A 6-bit ADC suffices for the hybrid.
- **Stuck cells are the limiting effect**: at 1% stuck cells every hybrid placement loses 26–40 points against its error-free accuracy, for every fault pattern, and even all-SLC loses 8–33. Contributions 3 and 4 address this.

2 Contribution 2: Fisher damage estimate and magnitude placement

What. The expected rise in loss if component r is stored in MLC, summed over every number it stores (its column $\sigma_r v_r$ in the first crossbar and its row u_r in the second):

$$D_r = s_r^2 \sum_j F(\sigma_r v_{rj}) + \sum_i t_i^2 F(u_{ir}),$$

where F is the empirical Fisher information (loss sensitivity) of each stored number and s_r, t_i are the 8-bit column scales that turn a fixed cell noise into a weight error. D_r is the principled estimate of what placement should minimise. We compare it with the HyFlexPIM gradient and with plain magnitude $|\sigma_r|$.

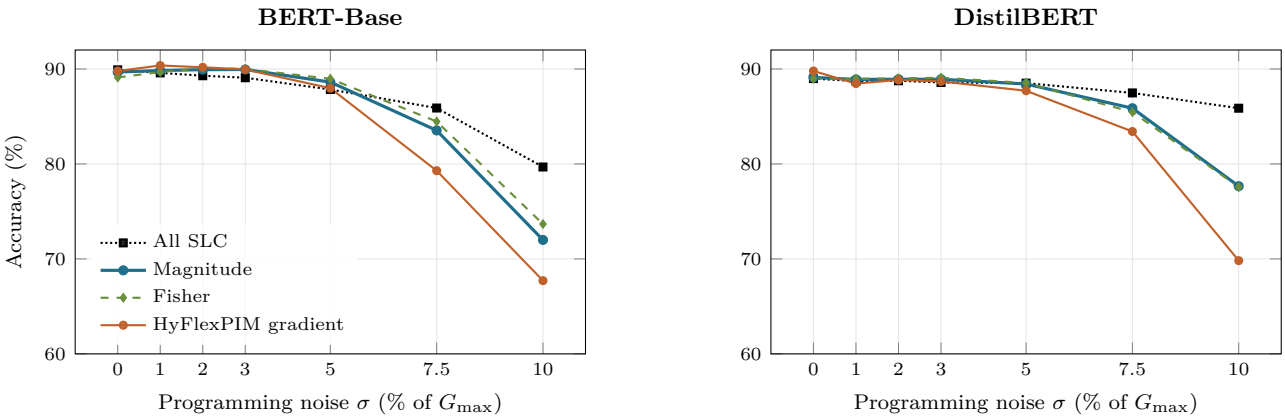


Figure 1: Accuracy against programming noise, 20% SLC, mean of 5 chips.

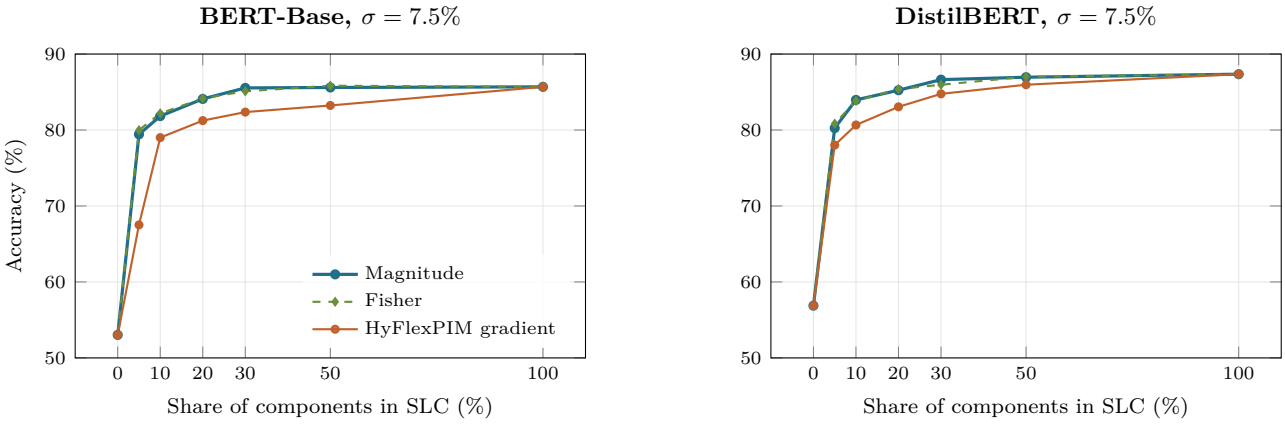


Figure 2: Accuracy against the share of components in SLC at $\sigma = 7.5\%$, mean of 20 chips; 0% is all-MLC, 100% all-SLC.

Table 2: Paired difference to magnitude, in correctly classified sentences out of 872 (mean ± standard error; 100 chips per noise level, 300 overall, all SLC shares).

Model	Placement	$\sigma = 3\%$	5%	7.5%	All
BERT-Base	Fisher	-0.1 ± 0.5	-0.1 ± 0.7	$+1.4 \pm 1.9$	$+0.4 \pm 0.7$
	HyFlexPIM gradient	-1.2 ± 0.7	-10.4 ± 1.6	-40.4 ± 5.0	-17.3 ± 2.0
DistilBERT	Fisher	-0.4 ± 0.7	-1.3 ± 0.7	$+0.0 \pm 1.7$	-0.6 ± 0.7
	HyFlexPIM gradient	-0.9 ± 0.7	-4.0 ± 1.0	-18.4 ± 2.6	-7.8 ± 1.1

Table 3: Smallest SLC share within half a point of all-SLC (shares tested: 5, 10, 20, 30, 50%), and the resulting cells per weight relative to all-SLC.

Model	Placement	$\sigma = 3\%$	$\sigma = 5\%$	$\sigma = 7.5\%$
BERT-Base	Magnitude	**5%** (59% of cells)	**5%** (59%)	**30%** (70%)
	HyFlexPIM gradient	5% (59%)	10% (61%)	not reached at 50%
DistilBERT	Magnitude	5% (59%)	**10%** (61%)	**50%** (79%)
	HyFlexPIM gradient	5% (59%)	20% (66%)	not reached at 50%

Takeaways.

- **Fisher and magnitude choose the same components**: rank correlation 0.96–0.97 on MNLI, and a paired difference within about one sentence in 872 at every noise level on SST-2 (Table 2).
- **Both beat the HyFlexPIM gradient at high noise**: by $+2.8 \pm 0.8$ and $+2.2 \pm 0.3$ points at 20% SLC and $\sigma = 7.5\%$, and by $+11.9 \pm 1.7$ points on BERT-Base at 5% SLC.
- **Why**: the column of component r holds $\sigma_r v_r$, so its scale, and the weight error a cell noise causes, is proportional to $|\sigma_r|$. Fisher multiplies this error size by the sensitivity; that it agrees with magnitude suggests the error size dominates. The gradient ignores the error size.
- **Practical result**: magnitude needs no data or gradients and half the SLC share of the gradient at $\sigma = 5\%$ (Table 3).

3 Contribution 3: fault-aware mapping

What. The components of an SVD layer can be stored in any order. After a chip is tested, each component is placed in the slot whose stuck cells hurt it least: the cost of component r in slot s is the Fisher-weighted squared error that slot's stuck cells would cause, and the Hungarian algorithm finds the best assignment. It needs no extra hardware or retraining. Joint mapping, which also chooses the tier per chip, is tested as an extension.

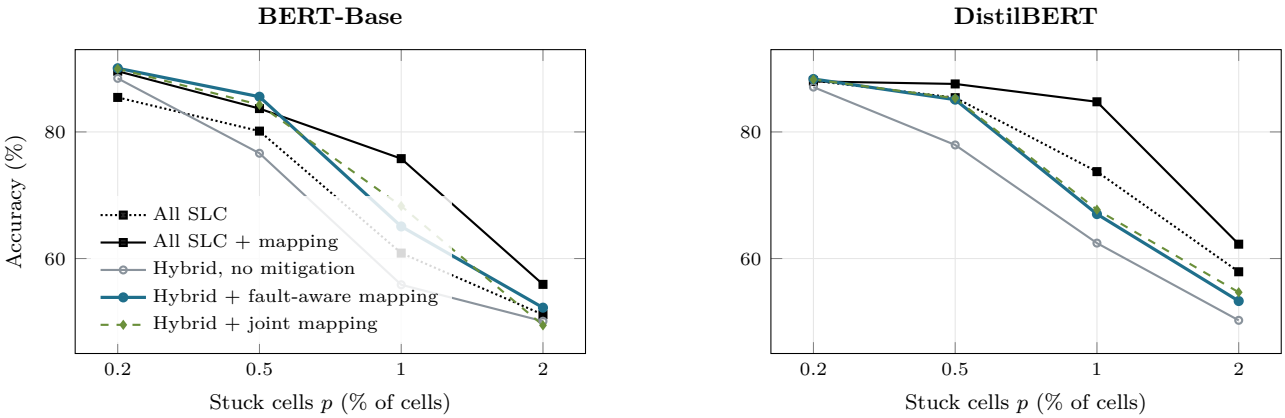


Figure 3: Accuracy against the share of stuck cells, no programming noise, mean of 10 chips. The hybrid uses magnitude placement with 20% SLC.

Table 4: Paired gain of fault-aware mapping over no mitigation (hybrid, magnitude, 20% SLC), and of joint tier-and-slot mapping over fault-aware mapping, in accuracy points (mean ± s.e. over 10 chips).

		Stuck cells, no noise				Stuck cells, $\sigma = 5\%$			1% stuck	
	Model	0.2%	0.5%	1%	2%	0.2%	0.5%	1%	clusters	lines
Mapping gain	BERT	$+1.6 \pm 0.3$	$+8.9 \pm 2.0$	$+9.2 \pm 2.6$	$+2.1 \pm 1.2$	$+2.5 \pm 0.6$	$+10.2 \pm 2.1$	$+6.0 \pm 1.7$	$+10.6 \pm 2.3$	$+6.8 \pm 2.2$
	Distil	$+1.3 \pm 0.5$	$+7.2 \pm 3.3$	$+4.6 \pm 3.4$	$+3.1 \pm 1.1$	$+2.3 \pm 0.5$	$+8.1 \pm 1.2$	$+5.6 \pm 3.5$	$+12.6 \pm 2.1$	$+13.9 \pm 3.3$
Joint − mapping	BERT	-0.1 ± 0.2	-1.3 ± 0.8	$+3.2 \pm 2.2$	-2.8 ± 0.9	-0.3 ± 0.3	-2.5 ± 1.9	$+3.6 \pm 2.0$	-2.8 ± 2.7	-0.1 ± 1.9
	Distil	-0.1 ± 0.1	$+0.2 \pm 0.7$	$+0.7 \pm 2.0$	$+1.4 \pm 1.6$	-0.7 ± 0.4	-0.7 ± 0.8	-2.0 ± 3.1	$+7.0 \pm 2.3$	-5.3 ± 2.4

Takeaways.

- **Mapping helps in every setting**, by 1.3–13.9 points; at 0.5% stuck cells BERT-Base rises from 76.6% to 85.6%, and the spread over chips falls from 6.7 to 1.2 points. Gains are largest for clusters and broken lines.
- **Joint mapping adds nothing reliable**: its second-order cost underrates the large error of a stuck high-order cell, so the benefit comes from the choice of slot, not of tier.
- **Mapping alone falls behind all-SLC from 0.5% stuck cells**, which contribution 4 closes.

4 Contribution 4: tier-aware training

What. Fine-tuning for 2 epochs on a freshly simulated chip at every step (5% noise, 0.5% stuck cells); the loss is measured on the damaged weights and the update applied to the clean ones (straight-through). Tier-

aware training simulates the real hybrid layout, so each component trains under the errors of its own tier; the tier-unaware variants put every component on SLC cells, with noise only or with noise and stuck cells.

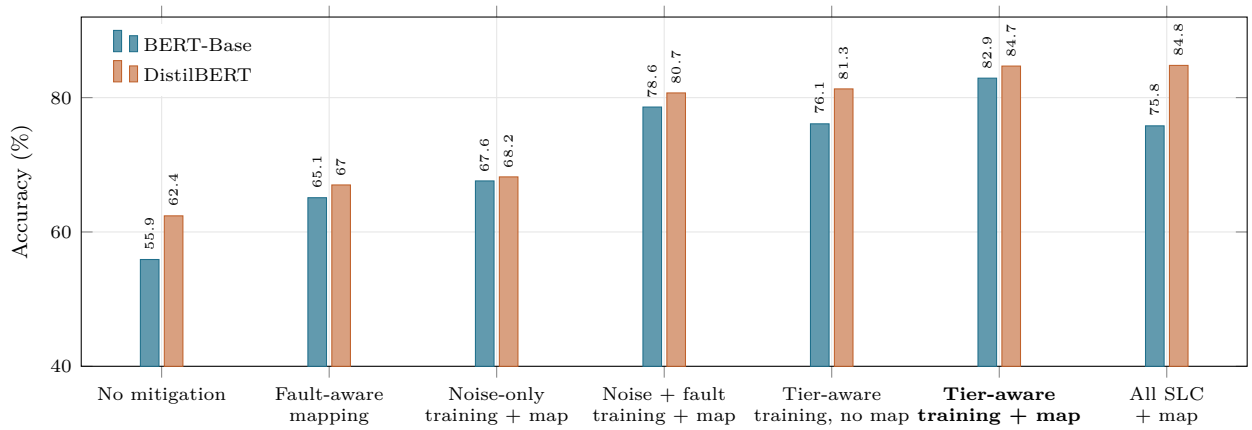


Figure 4: Accuracy at 1% stuck cells, no programming noise, mean of 10 chips. All bars except the last are the hybrid chip (magnitude placement, 20% SLC).

Table 5: Paired differences of tier-aware training + mapping, in accuracy points (mean ± s.e., 10 chips).

		Stuck cells, no noise				Stuck cells, $\sigma = 5\%$			1% stuck	
Compared with	Model	0.2%	0.5%	1%	2%	0.2%	0.5%	1%	clusters	lines
mapping only	BERT	-1.2 ± 0.2	$+2.7 \pm 0.5$	$+17.8 \pm 2.2$	$+0.8 \pm 1.0$	$+1.6 \pm 0.5$	$+7.4 \pm 0.8$	$+22.4 \pm 1.8$	$+20.7 \pm 1.6$	$+20.8 \pm 2.8$
	Distil	-0.1 ± 0.2	$+3.0 \pm 0.8$	$+17.7 \pm 2.7$	$+4.1 \pm 2.4$	$+0.5 \pm 0.4$	$+3.9 \pm 0.6$	$+14.7 \pm 2.4$	$+14.9 \pm 2.4$	$+16.5 \pm 3.1$
noise + fault training	BERT	-1.0 ± 0.2	-0.4 ± 0.4	$+4.3 \pm 2.2$	$+2.2 \pm 1.0$	-0.6 ± 0.4	$+0.4 \pm 0.3$	$+7.5 \pm 1.8$	$+6.2 \pm 1.7$	$+5.1 \pm 3.7$
	Distil	-0.4 ± 0.2	-0.1 ± 0.2	$+4.0 \pm 0.6$	$+2.4 \pm 1.7$	-0.5 ± 0.3	$+0.5 \pm 0.3$	$+2.3 \pm 1.6$	$+2.7 \pm 1.3$	$+2.8 \pm 0.8$
all SLC + mapping	BERT	-0.8 ± 0.3	$+4.6 \pm 0.7$	$+7.1 \pm 3.3$	-2.9 ± 2.2	$+2.9 \pm 1.4$	$+5.4 \pm 1.0$	$+4.0 \pm 1.7$	$+3.7 \pm 1.4$	$+0.2 \pm 3.0$
	Distil	$+0.3 \pm 0.2$	$+0.5 \pm 0.5$	-0.0 ± 0.9	-4.9 ± 3.1	$+0.7 \pm 0.3$	$+0.1 \pm 0.3$	-0.9 ± 2.4	-2.2 ± 1.2	$+3.0 \pm 1.9$

Table 6: Accuracy (%) under programming noise alone (no stuck cells), hybrid with 20% SLC by magnitude, mean of 10 chips; $\sigma = 0$ is an error-free chip.

	BERT-Base			DistilBERT		
Training	$\sigma = 0$	5%	7.5%	0	5%	7.5%
No training	89.7	88.7	84.2	89.1	88.2	85.6
Noise-only training	89.9	89.7	86.3	87.4	87.6	85.6
Noise + fault training	**90.9**	**89.9**	**88.8**	**89.9**	**89.3**	87.6
Tier-aware training	89.4	89.3	88.3	87.8	88.4	**88.0**

Takeaways.

- **Stuck cells must be injected**: noise-only training gives no consistent gain under stuck cells.
- **Knowing the tiers helps where errors are dense**: tier-aware training beats tier-unaware training at 1% stuck cells and with noise, and is slightly behind at 0.2% (cost 0.3 / 1.3 points on an error-free chip).
- **Training and mapping are complementary**, and together **close the gap to all-SLC**: the hybrid matches or beats all-SLC with mapping up to 1% scattered stuck cells, with 34% fewer cells.

5 Contribution 5: generalisation to MNLI

Table 7: MNLI (three classes, chance 33.3%): paired results, in accuracy points (mean ± s.e.). Raw accuracies are in Appendix A.6.

	BERT-Base	DistilBERT
Fisher − magnitude, $\sigma = 5\%$, all SLC shares (100 chips)	-0.2 ± 0.2	-0.2 ± 0.1
HyFlexPIM gradient − magnitude, same chips	-3.4 ± 0.4	-0.5 ± 0.2
Fault-aware mapping gain at 0.5% stuck cells (10 chips)	$+20.1 \pm 3.0$	$+18.8 \pm 3.2$

Magnitude again matches Fisher and beats the gradient on BERT-Base, and fault-aware mapping recovers about 20 points at 0.5% stuck cells. MNLI also exposed that one 8-bit scale for the vector between a tier's two crossbars costs up to 11 points even without errors, so only placements with equal tier sizes are compared.

6 First-order ADC cost estimate

Savings elsewhere in this document are counted in memory cells. ADCs dominate the power of crossbar accelerators (58% of tile power and 31% of tile area in ISAAC [1]), so we also estimate the ADC energy of each configuration analytically. We count the conversions of Eq. (5) of the report: one per output column, weight slice, 128-row tile and input bit (7 magnitude bits), with the two cells of a differential pair subtracted before the ADC. Each SVD layer is counted as its two crossbars per tier, so the tiling of the second crossbar's k_τ rows is included. The energy per conversion comes from ISAAC's 8-bit SAR ADC at 32 nm, 1.2 GS/s, eight ADCs per array group at 16 mW [1], i.e. 1.67 pJ per conversion; ISAAC takes these values from Murmann's ADC survey [2]. ISAAC scales an ADC to other resolutions by scaling its reference buffer, memory and clock linearly with the number of bits and its capacitive DAC exponentially [1]. We do not know how the energy splits between these parts, so we report both bounds: linear ($\propto b$) and exponential ($\propto 2^b$) scaling. The resolution of each configuration is the smallest that kept it within about a point of the ideal ADC (Section 1): 6 bits for all-SLC and the hybrid, 7 bits for all-MLC.

Table 8: ADC conversions and ADC energy per token for the encoder linear layers. Energy range: linear to exponential scaling of the ADC with resolution; the 8-bit all-SLC chip is the reference.

Model	Configuration	ADC bits	Conversions per token	ADC energy per token (μJ)	Relative energy
BERT-Base	All SLC	8	33.0 M	54.9	1
	All SLC	6	33.0 M	13.7–41.2	0.25–0.75
	All MLC	7	18.8 M	15.7–27.5	0.29–0.50
	Hybrid, 20% SLC	6	23.0 M	9.6–28.8	0.17–0.52
DistilBERT	All SLC	8	16.5 M	27.5	1
	All SLC	6	16.5 M	6.9–20.6	0.25–0.75
	All MLC	7	9.4 M	7.8–13.7	0.29–0.50
	Hybrid, 20% SLC	6	11.5 M	4.8–14.4	0.17–0.52

- At the same 6-bit resolution the hybrid needs 30% fewer conversions, and so 30% less ADC energy, than all-SLC. The saving is a little below the 34% saving in cells because the hybrid's MLC tier splits the second crossbar into more 128-row tiles.
- All-MLC needs fewer conversions still, but needs a 7-bit ADC, which costs 1.2–2× more energy per conversion than 6 bits, and it fails under programming noise and stuck cells.
- The estimate covers ADC conversions only. It leaves out DACs, sample-and-hold, shift-and-add and the digital layers, assumes one ADC energy for every conversion, and uses a 32 nm design point. It ranks the configurations; it is not a measurement. ADC area scales in the same way per ADC, but the number of ADCs depends on the throughput target, which we do not fix.

7 Overall comparison with the baseline

Table 9: Accuracy (%) of the baseline and of our methods, BERT-Base / DistilBERT. All hybrids keep 20% of the components in SLC unless stated. Programming-noise rows use the same 20 chips for all columns; stuck-cell rows take the baseline from the stuck-cell and fault-pattern sweeps and our methods from the mapping experiment (10 chips), so those comparisons are not paired.

Scenario	HyFlexPIM baseline	Magnitude	+ fault-aware mapping	+ tier-aware training	All SLC + mapping
$\sigma = 7.5\%$, 20% SLC	81.2 / 83.1	**84.1 / 85.2**	–	–	85.7 / 87.4[†]
$\sigma = 7.5\%$, 5% SLC	67.5 / 78.0	**79.4 / 80.2**	–	–	85.7 / 87.4[†]
0.5% stuck cells	71.0 / 74.5	76.6 / 77.9	85.6 / 85.1	**88.3 / 88.1**	83.7 / 87.6
1% stuck cells	52.8 / 55.4	55.9 / 62.4	65.1 / 67.0	**82.9 / 84.7**	75.8 / 84.8
1% stuck, 3×3 clusters	51.5 / 56.4	53.7 / 55.9	64.3 / 68.5	**85.0 / 83.4**	81.3 / 85.6
1% stuck, bit lines	51.0 / 52.2	50.1 / 54.0	56.9 / 68.0	**77.7 / 84.4**	77.5 / 81.5

[†]all-SLC without mapping (mapping has no effect without stuck cells).

Summary. Against the HyFlexPIM baseline, magnitude placement (justified by the Fisher estimate) removes the high-noise loss, fault-aware mapping recovers most of the stuck-cell loss at no hardware cost, and tier-aware training lifts the hybrid at 1% stuck cells from near chance to the level of an all-SLC chip, with a third fewer cells; the placement and mapping results also hold on MNLI. At the same ADC resolution the hybrid also needs about 30% less ADC energy than all-SLC.

References

[1] A. Shafiee, A. Nag, N. Muralimanohar, R. Balasubramonian, J. P. Strachan, M. Hu, R. S. Williams, and V. Srikumar, "ISAAC: A convolutional neural network accelerator with in-situ analog arithmetic in crossbars," in *Proc. 43rd Annu. Int. Symp. Comput. Archit. (ISCA)*, 2016, pp. 14–26.

[2] B. Murmann, "ADC performance survey 1997–2015 (ISSCC & VLSI Symposium)," 2015. [Online]. Available: `http://web.stanford.edu/~murmann/adcsurvey.html`

Appendix: complete results of our methods

Every result of our methods, both models and every sweep point, generated from the result files of the repository `github.com/Shivanshmittal023/reliable-reram-crossbar`, where each numbered step folder (`<step>`) stores `<step>/results/<experiment>/<model>/sweep_summary.csv`. The baseline configurations (all-SLC, all-MLC, random, HyFlexPIM gradient) are listed in the *Baseline Results* document and were measured on the same chips. Accuracy in %, mean over chips; hybrids keep 20% of the components in SLC unless stated.

A.1 Programming noise (5 chips per point)

Source: `02_programming_noise/results/noise/`.

Table 10: Magnitude and Fisher placement against programming noise σ.

Model	Placement	0%	1%	2%	3%	5%	7.5%	10%
BERT-Base	Magnitude	89.7	89.8	89.9	90.0	88.6	83.5	72.0
	Fisher	89.1	89.7	90.1	90.0	89.0	84.5	73.7
DistilBERT	Magnitude	89.1	88.9	88.9	88.9	88.4	85.9	77.7
	Fisher	89.0	89.0	89.0	89.1	88.5	85.5	77.6

A.2 SLC share (20 chips per point)

Source: `03_slc_share/results/slc_share/`.

Table 11: Magnitude and Fisher placement by share of components in SLC.

Model	σ	Placement	0% (all MLC)	5%	10%	20%	30%	50%	100% (all SLC)
BERT-Base	3%	Magnitude	82.1	89.7	89.9	89.8	89.8	89.7	89.2
		Fisher	82.1	89.6	89.8	89.8	89.8	89.8	89.2
	5%	Magnitude	66.8	88.0	88.6	88.7	88.9	88.9	88.0
		Fisher	66.8	87.9	88.6	88.8	88.9	88.9	88.0
	7.5%	Magnitude	53.0	79.4	81.8	84.1	85.5	85.6	85.7
		Fisher	53.0	80.0	82.2	84.2	85.1	85.8	85.7
DistilBERT	3%	Magnitude	86.0	88.4	88.6	88.8	88.9	88.7	88.7
		Fisher	86.0	88.5	88.7	88.7	88.8	88.6	88.7
	5%	Magnitude	74.8	87.4	88.0	88.1	88.8	88.4	88.2
		Fisher	74.8	87.3	87.8	88.4	88.1	88.3	88.2
	7.5%	Magnitude	56.9	80.2	84.0	85.2	86.6	86.9	87.4
		Fisher	56.9	80.8	83.8	85.4	86.0	87.0	87.4

A.3 Stuck cells (20 chips per point, no noise)

Source: `04_stuck_cells/results/fault/`.

Table 12: Magnitude and Fisher placement against the share of stuck cells p.

Model	Placement	0%	0.1%	0.2%	0.5%	1%	2%	5%
BERT-Base	Magnitude	89.7	89.6	88.6	76.8	57.9	51.0	50.0
	Fisher	89.1	89.7	88.4	75.6	55.5	49.5	50.3
DistilBERT	Magnitude	89.1	88.3	87.4	80.3	62.1	51.4	49.9
	Fisher	89.0	88.3	87.2	80.3	62.1	51.3	50.4

A.4 Other non-idealities

Source: `05_other_non_idealities/results/<experiment>/`.

Table 13: Read noise σ_r (5 chips).

Model	Placement	0%	0.5%	1%	2%	3%	5%
BERT-Base	Magnitude	90.6	90.1	89.9	90.1	89.3	82.5
	Fisher	89.6	90.0	89.9	89.7	89.0	82.7
DistilBERT	Magnitude	88.8	88.9	88.7	88.4	87.7	83.3
	Fisher	89.1	88.8	88.7	88.2	87.5	83.2

Table 14: MLC level-dependent noise at $\sigma = 5\%$ (10 chips).

Model	Placement	uniform	mid $\times 1.5$	mid $\times 2$
BERT-Base	Magnitude	88.9	88.7	88.4
	Fisher	89.0	88.9	88.4
DistilBERT	Magnitude	88.4	88.3	88.0
	Fisher	88.1	88.0	87.9

Table 15: Fault patterns at 1% stuck cells (10 chips).

Model	Placement	scattered	clusters	lines
BERT-Base	Magnitude	56.4	54.0	50.4
	Fisher	55.6	53.1	52.2
DistilBERT	Magnitude	62.5	56.1	54.7
	Fisher	60.0	57.3	56.3

Table 16: IR drop, wire resistance r (deterministic).

Model	Placement	0	10^{-4}	3×10^{-4}	10^{-3}	3×10^{-3}
BERT-Base	Magnitude	90.6	90.1	90.4	88.1	76.8
	Fisher	89.6	91.1	91.1	88.3	72.8
DistilBERT	Magnitude	88.8	89.2	89.7	86.6	73.4
	Fisher	89.1	89.2	89.3	86.4	72.6

Table 17: Fixed ADC with bit-serial inputs (deterministic).

Model	Placement	ideal	8 bit	7 bit	6 bit	5 bit	4 bit	3 bit
BERT-Base	Magnitude	90.6	90.4	89.3	90.0	51.1	59.9	52.6
	Fisher	89.6	90.1	89.7	90.5	51.6	57.0	52.9
DistilBERT	Magnitude	88.8	88.6	89.0	89.7	70.6	51.4	62.4
	Fisher	89.1	88.0	88.2	88.8	70.9	51.7	60.3

Table 18: Fixed ADC with $\sigma = 5\%$ programming noise (3 chips).

Model	Placement	ideal	8 bit	6 bit	4 bit
DistilBERT	Magnitude	88.7	89.1	88.5	51.5
	Fisher	87.8	88.5	88.2	52.3

A.5 Fault-aware mapping (10 chips per point)

Source: `06_fault_aware_mapping/results/{fault,fault_noise,pattern}/`.

All rows of this experiment, including the unmapped references, were measured on the same 10 chips.

Table 19: Stuck cells, no noise.

Model	Placement	0.2%	0.5%	1%	2%
BERT-Base	All SLC	85.4	80.1	60.9	51.1
	All SLC + fault-aware mapping	89.6	83.7	75.8	55.9
	Hybrid, no mitigation	88.5	76.6	55.9	50.1
	Hybrid + fault-aware mapping	90.1	85.6	65.1	52.2
	Hybrid + joint mapping	90.0	84.3	68.3	49.4
DistilBERT	All SLC	88.1	85.4	73.7	57.9
	All SLC + fault-aware mapping	88.0	87.6	84.8	62.3
	Hybrid, no mitigation	87.1	77.9	62.4	50.2
	Hybrid + fault-aware mapping	88.4	85.1	67.0	53.3
	Hybrid + joint mapping	88.3	85.3	67.7	54.7

Table 20: Stuck cells with $\sigma = 5\%$.

Model	Placement	0.2%	0.5%	1%
BERT-Base	All SLC	85.6	78.4	64.6
	All SLC + fault-aware mapping	86.1	82.4	76.5
	Hybrid, no mitigation	84.9	70.3	52.1
	Hybrid + fault-aware mapping	87.4	80.4	58.1
	Hybrid + joint mapping	87.2	77.9	61.7
DistilBERT	All SLC	86.9	81.7	68.0
	All SLC + fault-aware mapping	87.6	86.8	80.7
	Hybrid, no mitigation	85.4	74.9	59.5
	Hybrid + fault-aware mapping	87.7	83.0	65.1
	Hybrid + joint mapping	87.0	82.3	63.1

Table 21: 1% stuck cells by pattern.

Model	Placement	clusters	lines
BERT-Base	All SLC	66.5	57.2
	All SLC + fault-aware mapping	81.3	77.5
	Hybrid, no mitigation	53.7	50.1
	Hybrid + fault-aware mapping	64.3	56.9
	Hybrid + joint mapping	61.5	56.8
DistilBERT	All SLC	79.6	71.1
	All SLC + fault-aware mapping	85.6	81.5
	Hybrid, no mitigation	55.9	54.0
	Hybrid + fault-aware mapping	68.5	68.0
	Hybrid + joint mapping	75.5	62.7

A.6 MNLI (2,000 matched validation pairs)

Source: `08_mnli/results/{slc_share,fault,mapping}/`.

Table 22: MNLI: magnitude and Fisher placement by SLC share at $\sigma = 5\%$ (20 chips).

Model	σ	Placement	0% (all MLC)	5%	10%	20%	30%	50%	100% (all SLC)
BERT-Base	5%	Magnitude	33.5	64.0	66.0	68.0	69.2	66.7	61.6
		Fisher	33.5	64.2	65.8	67.8	68.8	66.6	61.6
DistilBERT	5%	Magnitude	36.8	68.4	70.0	71.5	72.6	72.5	72.0
		Fisher	36.8	68.5	70.0	71.1	72.1	72.3	72.0

Table 23: MNLI: magnitude and Fisher placement against stuck cells (20 chips).

Model	Placement	0%	0.1%	0.2%	0.5%	1%	2%	5%
BERT-Base	Magnitude	79.1	74.6	67.8	40.4	32.8	32.2	32.3
	Fisher	79.3	73.9	67.2	39.8	32.8	32.3	32.6
DistilBERT	Magnitude	77.3	74.8	71.3	49.7	34.0	33.3	33.8
	Fisher	78.3	74.9	70.7	50.3	34.0	33.6	33.1

Table 24: MNLI: fault-aware mapping (10 chips).

Model	Placement	0.2%	0.5%	1%	2%
BERT-Base	All SLC	61.5	46.7	35.7	33.2
	All SLC + fault-aware mapping	69.9	61.9	46.2	33.1
	Hybrid, no mitigation	68.1	40.0	33.1	32.2
	Hybrid + fault-aware mapping	75.3	60.0	36.5	32.3
	Hybrid + joint mapping	75.5	60.4	35.5	32.3
DistilBERT	All SLC	71.4	58.4	41.4	34.6
	All SLC + fault-aware mapping	76.2	72.2	58.5	37.4
	Hybrid, no mitigation	69.9	49.1	33.4	33.4
	Hybrid + fault-aware mapping	76.6	67.9	38.8	34.3
	Hybrid + joint mapping	76.6	67.0	39.4	33.3

A.7 Training (10 chips per point)

Source: `07_training/results/<variant>/<experiment>/`. Variants `nat_noise`, `nat_fault` and `tier_aware`; the untrained hybrid is in A.5.

Table 25: Hybrid chip (20% SLC by magnitude) for each training variant, without and with fault-aware mapping.

			Stuck cells, no noise				Stuck cells, $\sigma = 5\%$			1% stuck	
Model	Training	Mapping	0.2%	0.5%	1%	2%	0.2%	0.5%	1%	clusters	lines
BERT-Base	Noise only	no	89.3	80.6	56.9	51.0	87.1	74.1	52.9	54.4	50.1
		yes	89.8	87.0	67.6	51.7	88.0	82.7	58.0	65.8	58.9
	Noise + faults	no	89.9	87.5	66.3	50.3	88.6	83.7	60.2	58.0	52.2
		yes	89.9	88.7	78.6	50.8	89.7	87.4	72.9	78.8	72.6
	Tier-aware	no	89.2	87.1	76.1	50.8	88.6	86.5	70.0	71.3	53.1
		yes	88.8	88.3	82.9	53.0	89.1	87.8	80.5	85.0	77.7
DistilBERT	Noise only	no	86.3	77.5	62.1	50.5	85.3	76.4	60.6	56.6	53.6
		yes	87.3	84.5	68.2	51.5	85.9	82.8	59.9	66.1	65.9
	Noise + faults	no	88.6	83.4	72.5	50.8	88.1	83.9	67.7	68.0	57.6
		yes	88.6	88.2	80.7	55.0	88.8	86.4	77.5	80.7	81.6
	Tier-aware	no	88.2	85.7	81.2	52.9	88.1	85.6	76.5	77.7	67.5
		yes	88.2	88.1	84.7	57.4	88.2	86.9	79.8	83.4	84.4